

Chapter 5 Basics of Linear Algebra

5.1 Introduction

Linear Algebra is one of the fundamental mathematical disciplines used in Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Robotics, Natural Language Processing (NLP), and Data Science.

Modern AI systems process large amounts of numerical data. Images, text, audio, videos, and sensor measurements are all represented as vectors, matrices, or higher-dimensional arrays (tensors). Neural networks, recommendation systems, and Large Language Models (LLMs) rely heavily on linear algebra operations such as matrix multiplication, vector similarity, and transformations.

Understanding the basics of linear algebra helps learners appreciate how AI algorithms manipulate data internally.

5.2 Scalars

A **scalar** is a single numerical value.

Examples:

5, − 8, 3.14, 100

A scalar has only magnitude and no direction.

Examples of scalars include:

- Temperature
 - Age
 - Salary
 - Distance
 - Weight
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5.3 Vectors

A **vector** is an ordered collection of numbers arranged in one dimension.

A vector may be represented as a **row vector**

$$x = [2 \quad 5 \quad 8 \quad 10]$$

or as a **column vector**

$$x = \begin{bmatrix} 2 \\ 5 \\ 8 \\ 10 \end{bmatrix}$$

The number of elements in a vector is called its **dimension**.

Example:

$$x = \begin{bmatrix} 4 \\ 8 \\ 6 \end{bmatrix}$$

Dimension = 3

Real-world Example

Suppose a student scores the following marks:

Subject	Marks
Mathematics	85
Physics	91
Chemistry	88
Biology	79

The marks can be represented as

$$x = \begin{bmatrix} 85 \\ 91 \\ 88 \\ 79 \end{bmatrix}$$

5.4 Matrices

A **matrix** is a rectangular arrangement of numbers organized into rows and columns.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

The size of a matrix is represented as

Rows × Columns

The above matrix has

- 3 Rows
- 3 Columns

Hence,

$$A \in \mathbb{R}^{3 \times 3}$$

Example

$$B = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Dimension:

$$2 \times 3$$

5.5 Tensors

A **tensor** is a generalization of vectors and matrices to higher dimensions.

Object Dimension

Scalar 0D

Vector 1D

Matrix 2D

Tensor 3D or Higher

Examples:

- Grayscale Image $\rightarrow$ Matrix
 - Colour Image $\rightarrow$ 3D Tensor
 - Video $\rightarrow$ 4D Tensor
 - Mini-batch of Images $\rightarrow$ 4D Tensor
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5.6 Matrix Addition

Two matrices can be added only if they have the **same dimensions**.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Then,

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

5.7 Scalar Multiplication

Each element of the matrix is multiplied by a scalar.

Example

$$2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

5.8 Matrix Multiplication

Matrix multiplication is one of the most important operations in Artificial Intelligence.

Suppose

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}$$

Therefore,

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Condition for Matrix Multiplication

If

$$A_{m \times n}$$

and

$$B_{n \times p}$$

then

$$AB$$

is possible and the resulting matrix has dimension

$$m \times p$$

Example

$$(3 \times 2) \times (2 \times 4) = 3 \times 4$$

5.9 Transpose of a Matrix

The transpose interchanges rows and columns.

If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

then

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

5.10 Identity Matrix

An identity matrix is a square matrix having

- Ones on the principal diagonal
- Zeros elsewhere

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Property:

$$AI = A$$

5.11 Dot Product

The dot product is defined as

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

Example

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, y = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Then

$$1 \times 4 + 2 \times 5 + 3 \times 6 = 32$$

Hence,

$$x \cdot y = 32$$

5.12 Euclidean Distance

The Euclidean distance between two vectors is

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Example

For the points

$$(2, 3)$$

and

$$(5, 7)$$

Distance

$$= \sqrt{(5 - 2)^2 + (7 - 3)^2} = \sqrt{9 + 16} = 5$$